CATERPILLAR®

DEALER BEST PRACTICE SERIES

Component Life Improvement Process

Maintenance and Repair | Application | Component Renewal (CRC) | **Component Life Management** | MARC Management

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1.0 Introduction

Machine drive train major component rebuild costs account for over 60% of total machine maintenance and repairs costs excluding tires and fuel. The most effective way to reduce machine operating costs is through improving major component life. This directly impacts machine cost per hour, and therefore cost per ton.

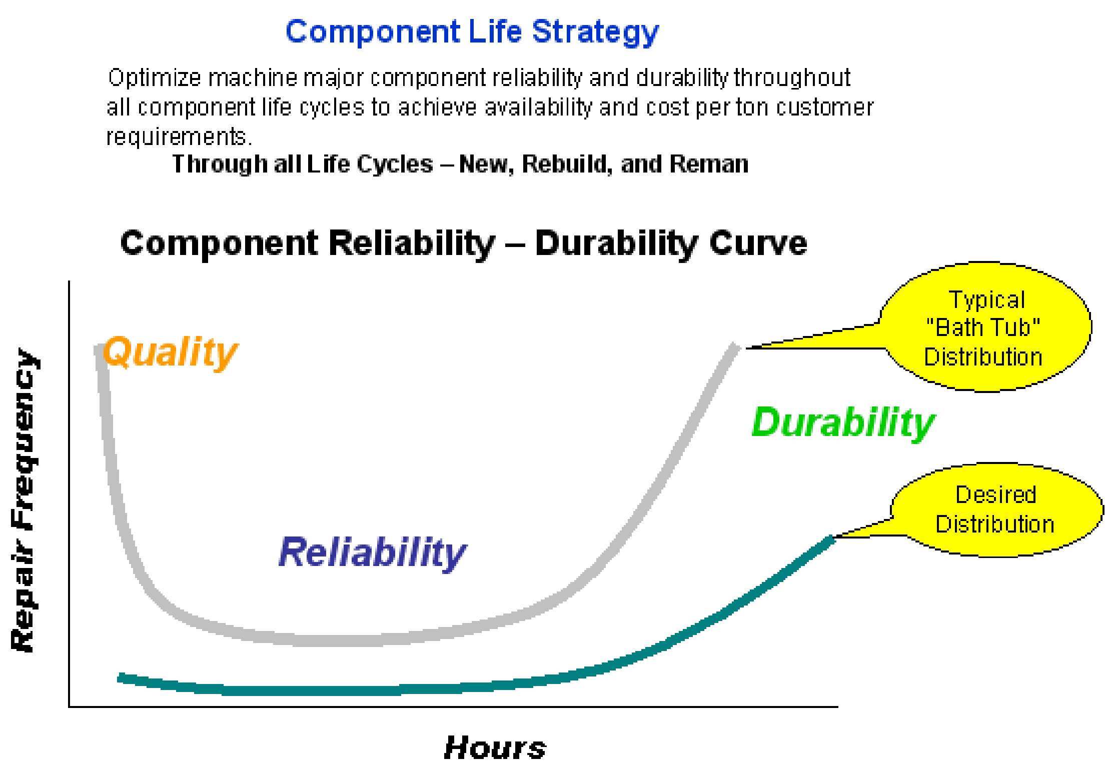
Improving major component life also has the additional benefit of reducing the frequency of planned and unplanned machine stoppages, which increases machine overall availability and production.

This Best Practice discusses the fundamentals and process for managing and improving major component life.

2.0 Best Practice Description

Managing major component life involves two basic fundamentals:

- Eliminate or reduce unplanned component removals due to failures or premature wear. Since failures often shorten life by 50% or more, actions to reduce the frequency of failures has a significant impact on average fleet component life.
- Set, monitor, evaluate, and adjust planned component replacement (PCR) targets for optimum life.



Process

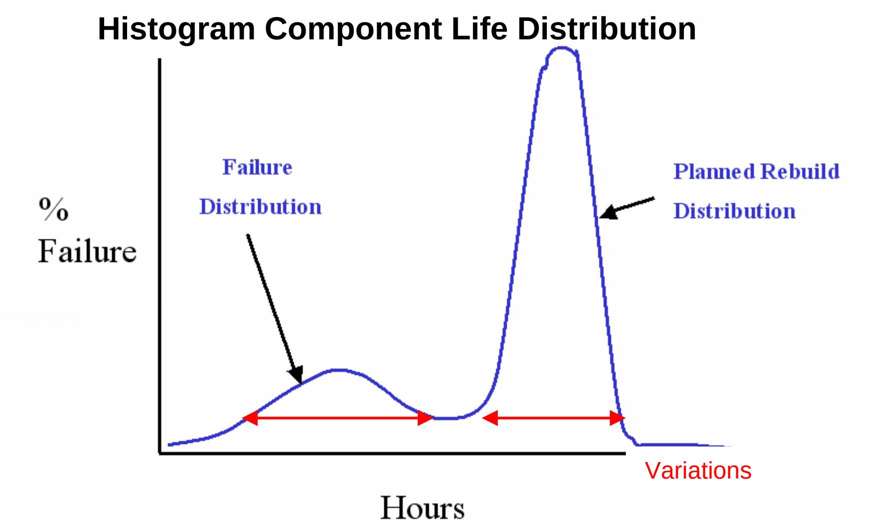
1. Collect and analyze component life data and information.

- Fleet component life hours achieved are tracked and summarized by machine model. Suggested required data includes:
  - Machine model
  - Machine serial number
  - Component name (such as engine, transmission, wheel group, differential, torque converter, suspension, etc)
  - Component serial number (or assigned ID number)
  - Component removal date.
  - Hours on component when removed.
  - Removal planned or unplanned? (Failure or Planned Component Replacement)
  - Root Cause of Failure. (Part / Part Group causing failure)
  - Component source (Component removed was New, Reman, or Rebuilt)
  - Other data as appropriate, such as "Total engine fuel burn" can be helpful.

- The component life data should be collected as a minimum for all component removals over the past three years. Over a three year period, close to 100% of the fleet components will have completed a life cycle. Component life data that is more than five years old should not be used in the analysis of "current" fleet component life performance.

- Useful data analysis tools are histogram and box plot graphs, which illustrate the distribution of the data and variation in the data. Data can be sorted and reviewed by any of the data fields above. (By model, by component, by new components, by rebuilt components, by failures, by PCRs, etc)

- A normal distribution often has two "bell curves". One curve represents failures and early removals - the other represents planned removals.



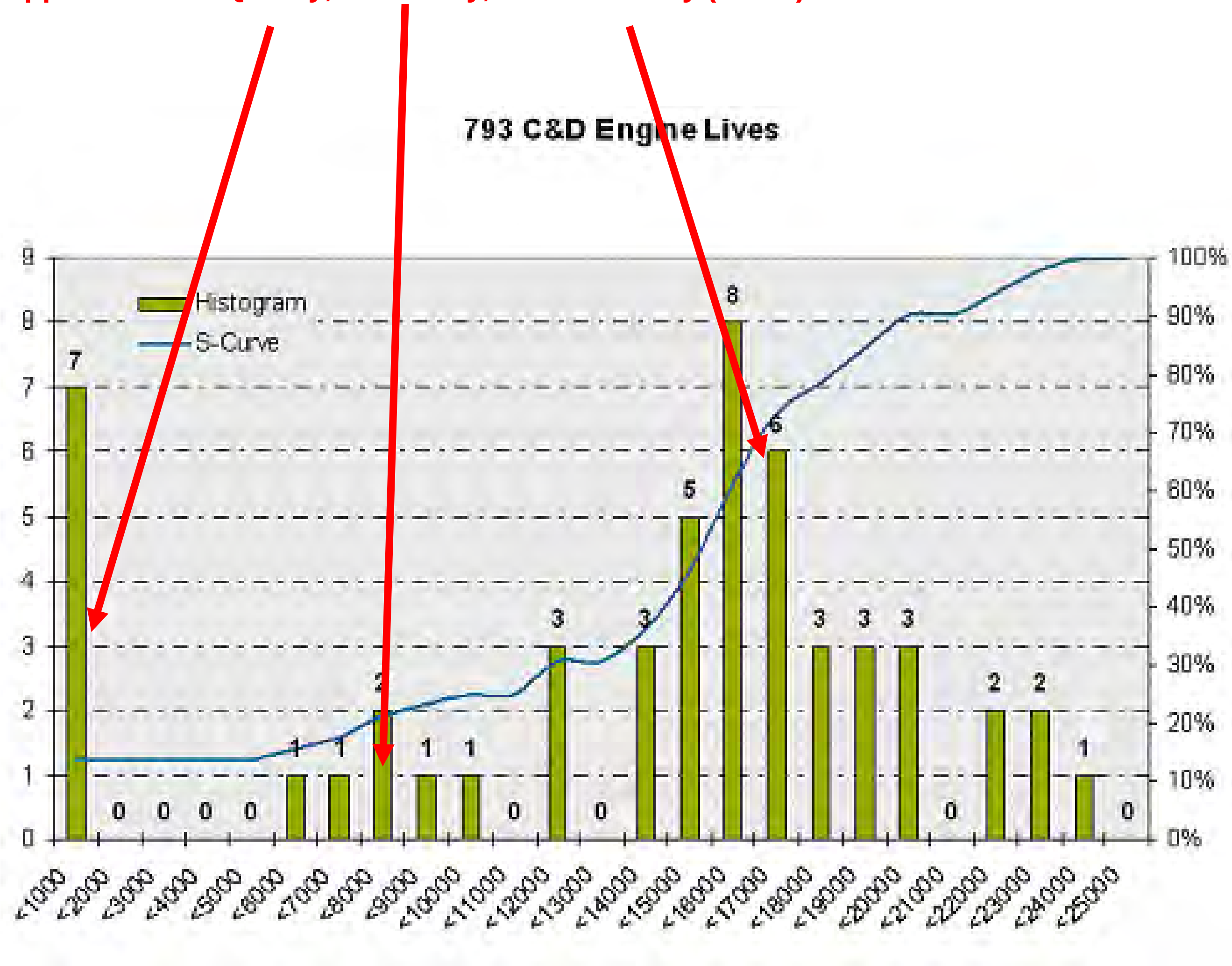
Histograms and S-curves:

This analysis tool shows the distribution by hours of all component removals (both failures and PCRs) and the accumulated percentage of the fleet that has completed a life cycle.

Note that the below actual data histogram shows a typical Reliability – Durability "Bath Tub" curve as shown in the "Strategy" histogram above.

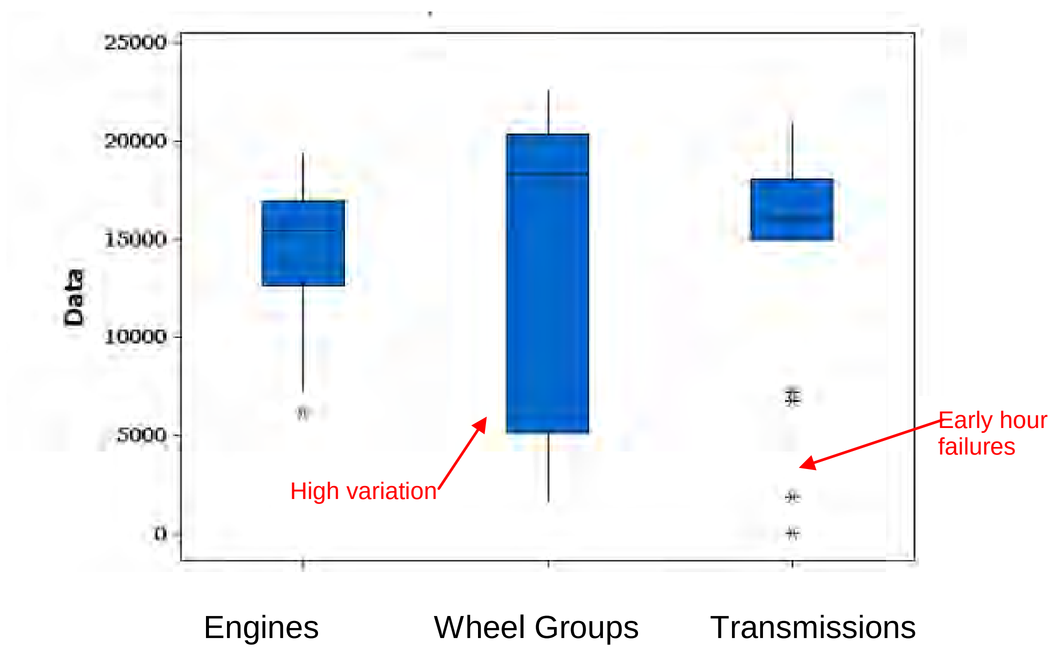
From the distribution of the removal hours and incidents, "opportunities" can be identified.

Opportunities: Quality, Reliability, and Durability (PCRs)

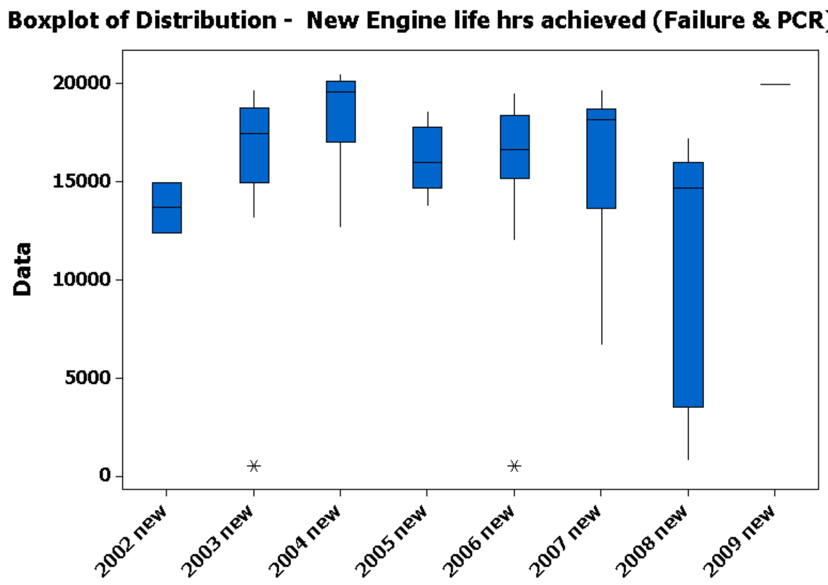


- Other data analysis tools.

793C Box Plot



Box Plot (by machine model) shows variation and outlaying data points.



Box Plot (by engines and by year) … shows trends and variations

2. Reducing Early Failures and Premature Wear Out

The management and reduction of early hour failures and premature wear out is a key success factor in increasing overall component life and machine reliability,

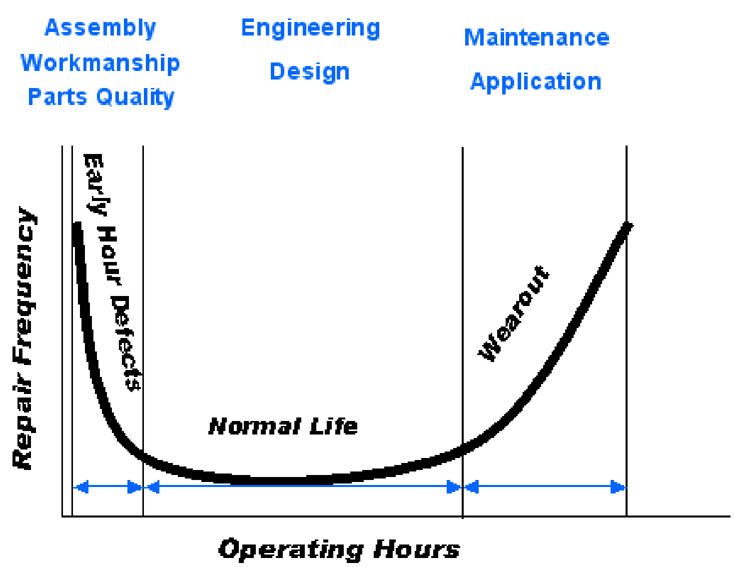
- Identify the root causes of failure impacting the machines.
- Determine and prioritize the top issues.
- Develop containment and solutions

Root Causes

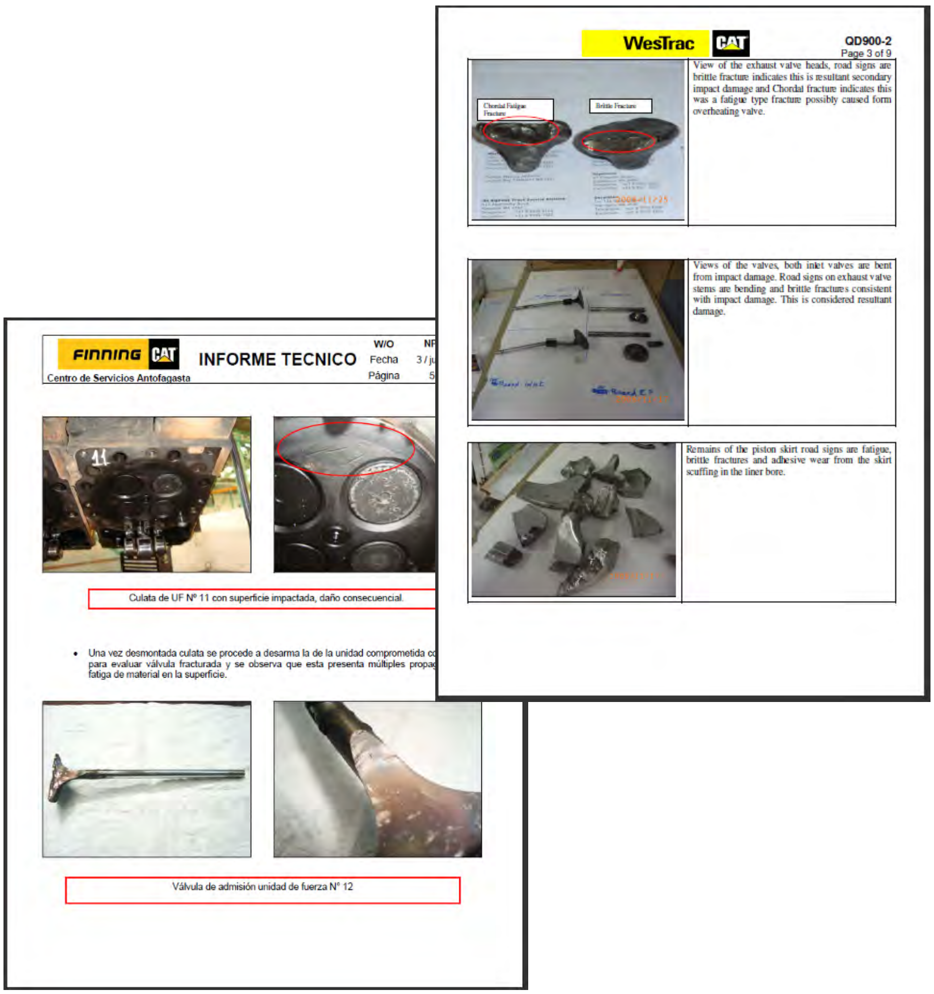
It is important to the process to identify the "root cause" for any component removed prior to reaching the planned life target. A technical failure analysis is the best means to identify the root causes of failures.
A common mistake is to identify "what" failed rather than "why" it failed. Understanding true root causes enables preventative actions to be developed and implemented.

Failures usually originate from one or more of the following sources:

- Part defect (Caterpillar or Rebuilder Reuse and Salvage)
- Workmanship defect (Caterpillar or Rebuilder)
- Engineering Design (Caterpillar)
- Maintenance practices (Minesite)
- Operation practices (Minesite)
- Improper Removal and Installation -system contamination (Installer)

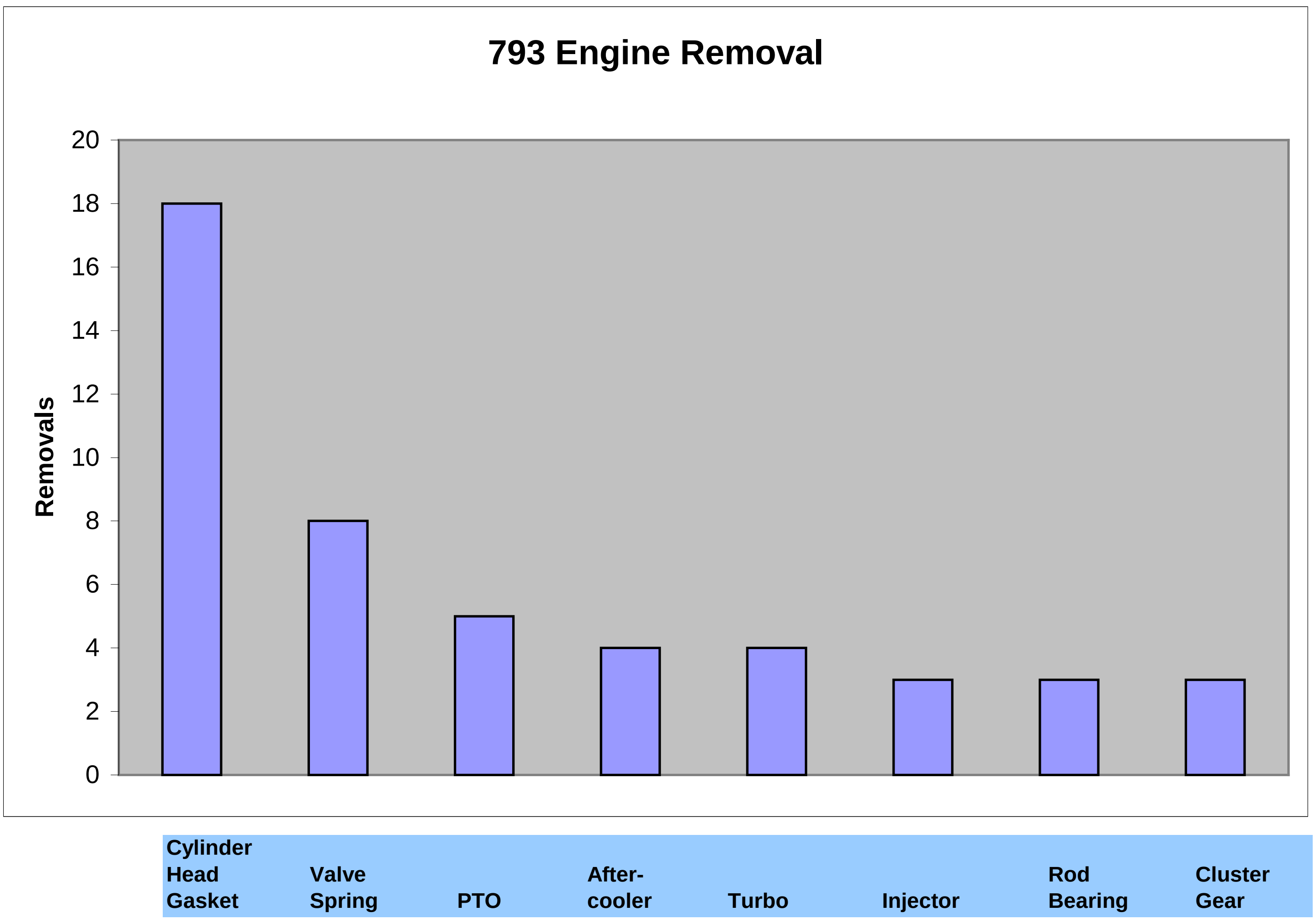


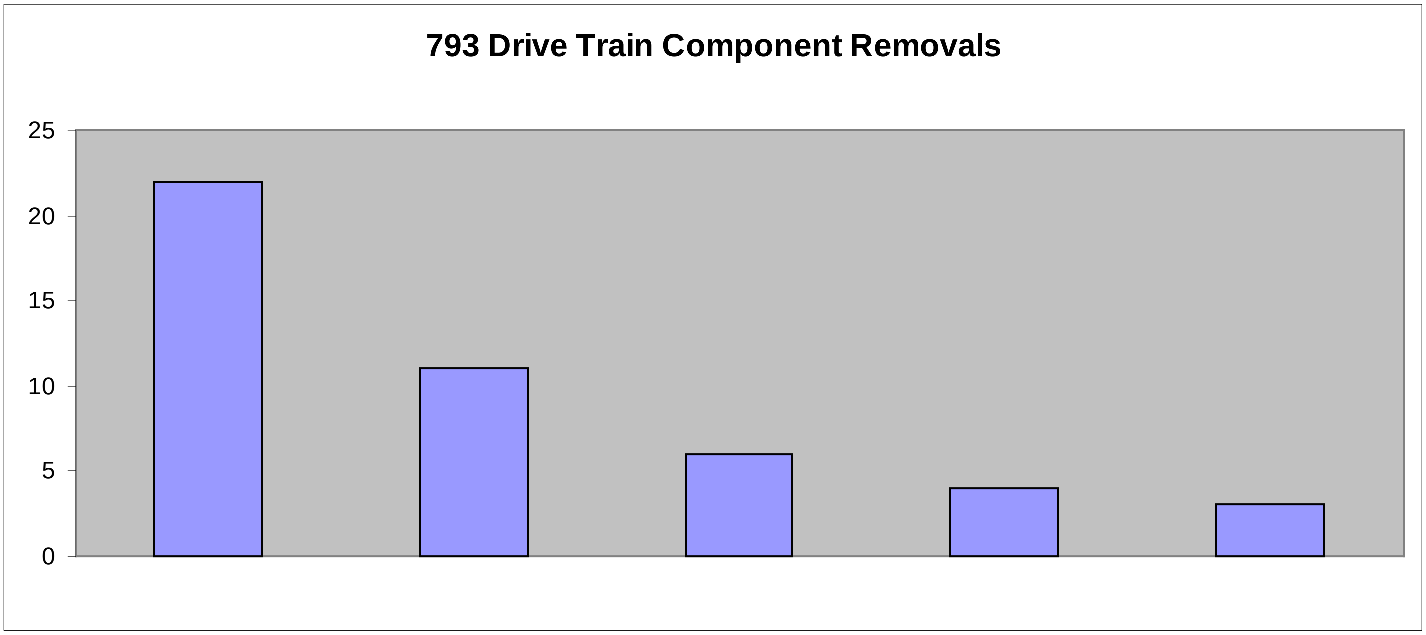
It is vital to establish a policy to perform technical failure analysis on all premature removals and a resource to provide the analysis. Most Caterpillar dealers have extensive Applied Failure Analysis (AFA) training, tooling and lab equipment required, as well as extensive knowledge of Caterpillar equipment, specifications, and technical product issues.



Define and Prioritize Issues

The accumulation of failure analysis information allows the top reasons for failures to be identified and prioritized by those issues which are most impacting component life and machine availability. Plans and actions can then be developed to contain known problems and find permanent solutions.
Pareto charts are often used to summarize and prioritize root causes of both downtime as well as major repairs that cause a component to be removed. Below is an example engine major repair pareto.





Steering Pump	Dou-cone seals	Rear Axle Lube Pump	Thrust button	Trans Control Valve

Action Plan – Containment and Solutions

Once key issues effecting component life performance and availability are identified and prioritized, action plans can be developed. Action plans consist of temporary containment measures and permanent corrective measures.

Temporary containment options may include:

- Identify which machines that may be affected by an issue.
- Inspect machines for a defect.
- Monitor machines for signs of a developing problem. VIMS data, oil analysis, particle count, magnetic plug and filter inspections, testing, etc.
- Perform a mid-life replacement of part or subcomponent known to be causing issues.
- Temporary machine systems modification.

Permanent solutions may include:

- Updated part or design from Caterpillar
- Change in workmanship procedure.
- Change in maintenance procedure.
- Change in operation procedure.

Note:

While the above process is to a method to manage “component life”, a similar process is often used to manage “availability”. Both involve identification of the top drivers and a containment and resolution plan. For component life management, measure is hours at removal. For availability management, measure is downtime, often measured as Mean Time Between Stoppage (MTBS) and Mean Time to Repair (MTTR).

Example of “Containment” Plan

Issue: 793D Dou-Cone Seals

- The Rear Wheel & Brake Group fails due to duo cone seal leaks. The toric rings have been reported to take a permanent set. This causes face load loss that results in a brake oil leak.
- Duo cone seal repairs accounted for 18 repairs and 582 repair hours in past 12 months.

Caterpillar Input:

- Aware of issue – improved seal released reference Service Magazine SEPD0971

Containment Action:

- Inspect for leaks at 500 hour PM
- Monitor oil analysis, viscosity, and particle count at 500 hour PM
- Identify machines running with old seals and those with new seals
- Remove old seal part numbers from parts stock
- Update component exchange inventory.

Implement Solution:

- Install new seal part number at rebuild or at repair.

Example of “Solution” Plan

Issue: Early Hours Transmission failures due to installation errors

- 7 low hour transmission failures for 238 repair hours in past 12 months.

Caterpillar Input:

- Use recommended R&I checklist and installation Special Instruction procedure

Implement Solution:

- Create a checklist for technicians.
- Supervisor must sign off on checklist
- Whenever a transmission is rebuilt, also rebuild the transmission pump and cooler.
- Replace all fluid lines during installation.
- Check control valve settings on commissioning.
- Always replace drive shaft bolts with new bolts – do not reuse bolts.
- Check fluid oil analysis and particle count after 24 hours – kidney lube oil system if needed.
- Develop a standard installation parts required “kit”.
- Training for technicians.

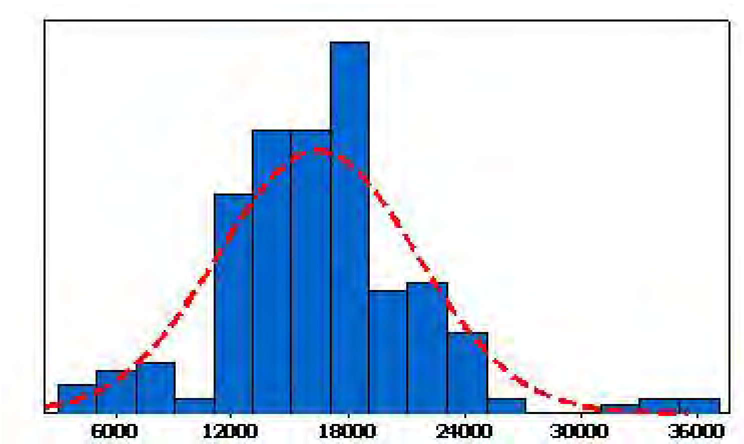
3. Manage and Adjust Planned Component Replacement Intervals

- Determine Component Life Targets
  - Cat / Dealer Recommendation
  - Experience and History
  - Experience in similar applications.
  - Load Factors – Fuel Burn Rate

- Track Component PCR Life
  - Average PCR Life
  - Life Histogram – Distribution and Variation
  - Percentage of Components Reaching PCR

- Adjusting PCR Targets
  - Evaluate Component Condition during rebuild – Life remaining?
  - Rebuild based on condition rather than by hours alone
    - Monitor oil consumption, smoke, oil analysis, particle count, VIMS, etc
  - Evaluate parts reusability/rebuild costs at PCR.

Histogram Wheel Group Life



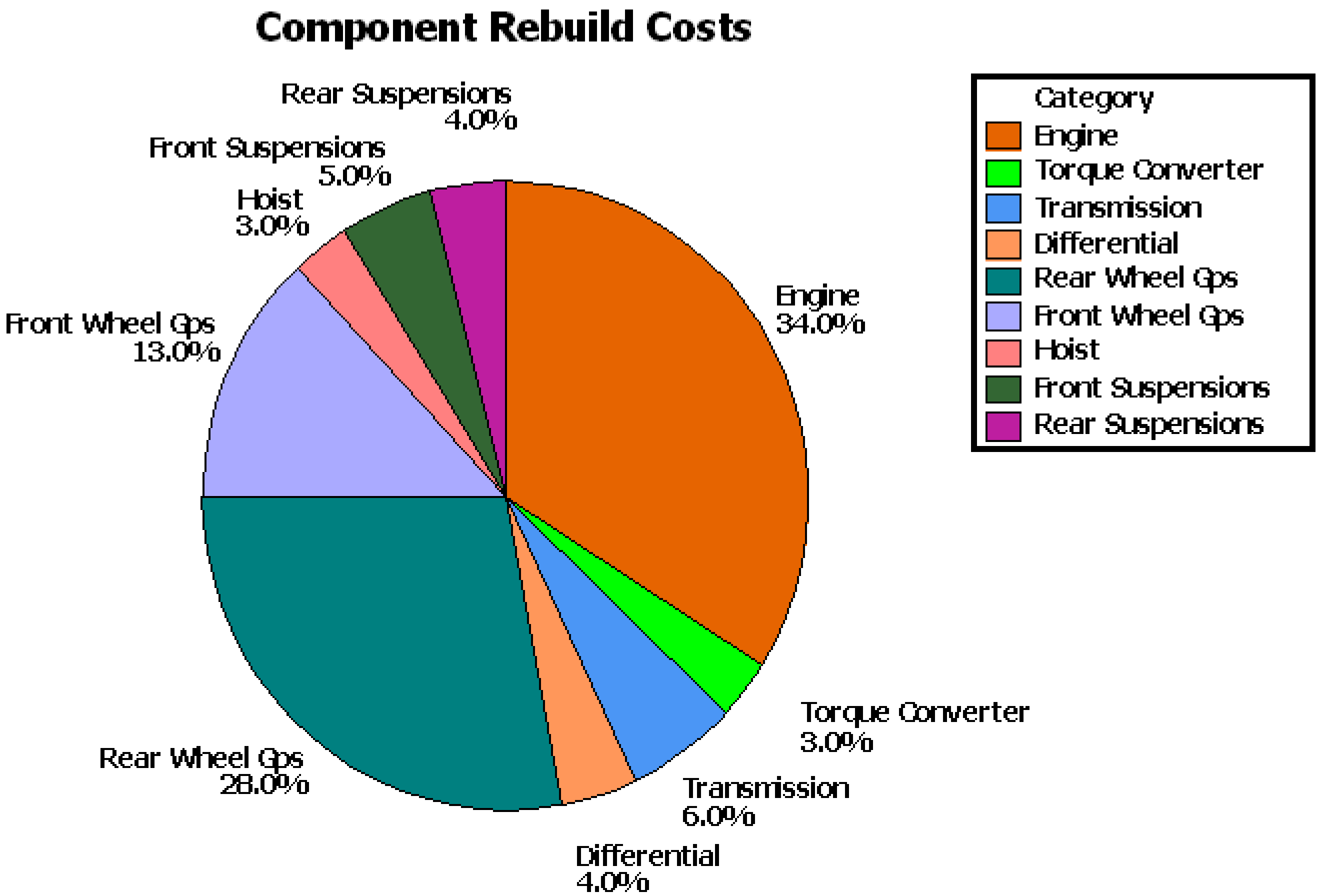
PCR Management Benchmarks

- **Variation -** PCRs are performed at +/- 5% of target hours. (Scheduling)
- **Distribution –** 80% of components reach PCR target. (Maint. / Monitoring)

Additional Information:

Cost Drivers

Typical rebuild cost comparison of various machine components show that for Off Highway Trucks, the engines and wheel groups are the key cost drivers and thus the key opportunity for cost reduction through improved life.



3.0 Implementation Steps

The fundamentals of this component life improvement process are:

- Collect and Analyze Component Life Data
- Manage Early Hour Failures and Premature Wear Out.
- Manage and Adjust Planned Component Rebuild (PCR) Intervals

To implement, the requirements are:

- Company commitment to create an availability / component life management process.
- Process owner: Assign a Reliability Manager / Engineer
- Administrative system and database to track component life.
- Maintenance and Repair records database
- Policy on performing failure analysis and a database for reports and history tracking.
- Developed desired management reports.
- Use data to identify opportunities.
- Explore opportunities for causes and solutions.
- Implement containment and solutions.

4.0 Benefits

- Improved fleet availability and production.
- Increased average fleet component life lowers cost per hour.
- Reduced unplanned machine stoppages – improved scheduling and planning.
- Lower parts consumption due to reduced failure rates.

5.0 Resources Required

- Staff
- Cost of AFA ‘s
- Cost of Databases

6.0 Supporting Attachments / References

Sample Component Life Tracking Spreadsheet

7.0 Related Best Practices

8.0 Acknowledgements